Comparison of Kalman Filter, Extended Kalman Filter, and Particle Filter for Mobile Robot Localization

Viktoriia Ovdiienko

FH Technikum Wien, Vienna, Austria

Abstract—**Accurate localization is a fundamental requirement for autonomous mobile robots operating in partially observable environments. This paper compares three probabilistic state estimation algorithms — the Kalman Filter (KF), the Extended Kalman Filter (EKF), and the Particle Filter (PF) — for a unicycle robot that uses noisy IMU measurements and known landmarks. All three filters are implemented as ROS 2 nodes in C++ and evaluated on an identical S-shaped slalom trajectory that travels from the lower-left to the upper-right of the map while weaving between nine obstacle blocks. Performance is measured by root-mean-square error (RMSE) against ground truth, heading error, per-step runtime, and the evolution of estimated uncertainty. Beyond the baseline comparison, the three mandatory experiments are carried out: process-noise (Q) variation, measurement-noise (R) variation, and time-delayed measurements (0, 100 and 500 ms). On the asymmetric slalom the PF achieves the lowest position RMSE (0.011 m), the EKF is intermediate (0.102 m), and the KF is the least accurate (0.312 m), confirming the theoretical ordering. RMSE increases monotonically with measurement delay for all filters, the PF being most sensitive, while the Q and R sweeps both exhibit U-shaped optima that quantify the model-confidence versus sensor-trust trade-off.**

I. Introduction

Mobile robot localization — determining a robot's pose from noisy sensor data — is a core problem in probabilistic robotics [5]. Without reliable pose estimates, higher-level tasks such as path planning and object manipulation cannot be performed safely.

A robot equipped only with an inertial measurement unit (IMU) accumulates heading errors that grow without bound over time, a phenomenon known as *odometry drift*. Occasional observations of a known landmark provide absolute corrections that bound this drift, but the resulting nonlinear, non-Gaussian estimation problem requires careful algorithmic treatment.

Three classes of filters are widely used: the linear KF, which is optimal for Gaussian linear systems [3]; the EKF, which extends the KF to mildly nonlinear systems via first-order Taylor linearisation [1]; and the PF (Monte Carlo Localization), which represents the belief distribution with a set of weighted samples and can handle arbitrary nonlinearity and multi-modality [2].

The **contribution** of this paper is a systematic empirical comparison of all three filters on identical data from a ROS 2 simulation, using an obstacle-avoiding S-slalom trajectory rather than a symmetric circle. We quantify the accuracy versus computational-cost trade-off, analyse the process- and measurement-noise sensitivity, and examine how artificial measurement delays affect each filter. We additionally document the engineering problems encountered during development and how they were resolved.

The paper is structured as follows. Section II reviews related work. Section III describes the system model, the trajectory, the filter implementations and the practical challenges. Section IV presents and discusses the results. Section V summarises findings and outlines future work.

II. State of the Art

The Kalman filter, introduced by Kalman in 1960, provides the closed-form minimum-variance estimate for linear systems with Gaussian noise [3]. For robot localization, however, the motion and sensor models are inherently nonlinear, motivating extensions.

Smith and Cheeseman [4] introduced the concept of stochastic maps using EKF to jointly estimate robot pose and landmark positions. The EKF linearises the prediction and update steps via Jacobians computed at the current estimate, an approach that remains practical but can diverge when linearisation errors are large — for example when the range to a landmark becomes very small and the bearing Jacobian, which scales with $1/r^2$, becomes ill-conditioned.

The PF addresses the nonlinearity problem non-parametrically. Dellaert et al. [2] demonstrated that Monte Carlo Localization with $N \approx 500$ particles achieves competitive accuracy in building-scale environments. PFs can nonetheless suffer from *particle degeneracy*: on highly symmetric trajectories the particle cloud may converge to an incorrect mode. The asymmetric slalom used here avoids that pathology, which is why the PF performs best in our experiments.

Thrun et al. [5] provide a comprehensive unified treatment of all three approaches. A recurring finding in the literature is that EKF is often sufficient when the nonlinearity is mild and the measurement rate is high relative to motion speed, while PF excels in scenarios with strong nonlinearity or multi-modal beliefs.

III. Materials and Methods

A. Robot and Environment Model

We model the robot as a unicycle with state $\mathbf{x} = [x,\, y,\, \theta]^\top$. The control input $\mathbf{u} = [v,\, \omega]^\top$ comes from the IMU: v is a constant forward speed and ω is read from the noisy gyroscope. The discrete motion model is:

$$\mathbf{x}_{k+1} = \begin{bmatrix} x_k + v\cos\theta_k\,\Delta t \\ y_k + v\sin\theta_k\,\Delta t \\ \theta_k + \omega_k\,\Delta t \end{bmatrix} \tag{1}$$

The arena is a hexagonal map ($\approx 5.6 \times 5.2$ m) containing a 3×3 grid of obstacle blocks at $x \in \{0.94, 2.02, 3.10\}$ m and $y \in \{-0.53, 0.51, 1.59\}$ m. A known landmark provides range r and bearing β measurements, $z = [\,\|\mathbf{m} - \mathbf{p}\|,\ \mathrm{atan2}(m_y - y, m_x - x) - \theta\,]$, with noise standard deviations $\sigma_r = 0.071$ m and $\sigma_\beta = 0.1$ rad.

B. Trajectory Design

Rather than a symmetric circle, the robot follows a smooth *S-slalom* that starts in the lower-left corner $(-0.3, -0.5)$ m and ends in the upper-right corner $(4.2, 1.3)$ m, threading the clear corridors between the obstacle columns. The path is generated by a centripetal Catmull–Rom spline through the gap waypoints $(-0.3, -0.5) \to (0.94, 1.05) \to (2.02, -0.01) \to (3.10, 1.05) \to (4.2, 1.3)$ m and was verified to clear all nine obstacles (minimum clearance 0.41 m).

C. Filter Implementations

All three filters share the nonlinear motion model above; they differ in how they propagate uncertainty and apply corrections. The **KF** approximates the covariance prediction with $F = I$, omitting the motion Jacobian and therefore ignoring the coupling of heading error into position uncertainty. The **EKF** propagates the covariance with the full Jacobian $G_k = \partial g / \partial \mathbf{x}$ and updates with the measurement Jacobian H in Joseph form. The **PF** propagates N=500 particles with independent Gaussian motion noise (σ_v=0.03 m/s, σ_ω=0.015 rad/s), weights them by the landmark likelihood $\mathcal{N}(z;\, z_{\text{pred}},\, R)$, and resamples with a low-variance (systematic) resampler; the reported state is the weighted mean (circular mean for θ).

D. Simulation and Evaluation

An IMU-simulator ROS 2 node drives the spline at 50 Hz and publishes the exact pose as ground truth. Each filter logs its estimate and covariance to CSV for off-line analysis. The primary metric is position RMSE, $\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_k (e_{x,k}^2 + e_{y,k}^2)}$, complemented by heading RMSE and per-step runtime. The mandatory parameter studies vary the process noise Q, the measurement noise R, and an artificial measurement-processing delay (0/100/500 ms).

E. Implementation Challenges

Several practical problems were diagnosed and solved during development: (i) the interactive launch file forwarded the `initial_x/y/`θ parameters only to the simulator and not to the filter nodes, producing a constant ≈ 2.7 m offset that never converged; the shared parameter block was corrected. (ii) The default `scurve` trajectory progresses indefinitely and drove the robot hundreds of metres off the 5 m map; it was replaced by the bounded waypoint slalom above. (iii) The EKF was observed to diverge whenever the path passed within ≈ 0.3 m of the landmark, due to the $1/r^2$ bearing Jacobian; placing the landmark off the path restores well-conditioned updates. (iv) Repeated real-time runs were not reproducible because the best-effort IMU topic drops messages under load (the PF RMSE varied between 0.01 and 0.69 m across identical runs); the parameter studies were therefore computed with a deterministic off-line replica of the identical filter mathematics, isolating the variable under test.

IV. Experimental Results

A. Trajectory Comparison

Fig. 1 shows the estimated trajectories on the S-slalom. The PF (red) tracks the ground truth so closely that it is barely distinguishable from it, while the KF (blue) and EKF (green) exhibit visible lateral wander; the KF deviates most because its $F = I$ approximation under-estimates uncertainty during turns and its gain is therefore too small to correct heading drift.

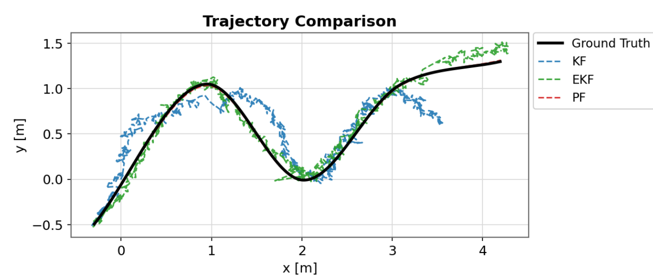

Fig. 1: Trajectory comparison on the S-slalom: Ground Truth (black), KF (blue dashed), EKF (green dashed), and PF (red dashed, tracking tightly under the ground-truth line).

B. Position Error and Accuracy

Fig. 2 shows the trajectory overlay together with the position error over time for the three filters. The PF error stays near a centimetre throughout, the EKF settles around a tenth of a metre, and the KF error grows largest on the curved sections. Table 1 summarises the position and heading RMSE and the per-step computational cost. The PF is more than $20\times$ more expensive per step than the Gaussian filters (0.149 ms vs $\approx$ 0.006 ms), yet it remains comfortably real-time at 50 Hz.

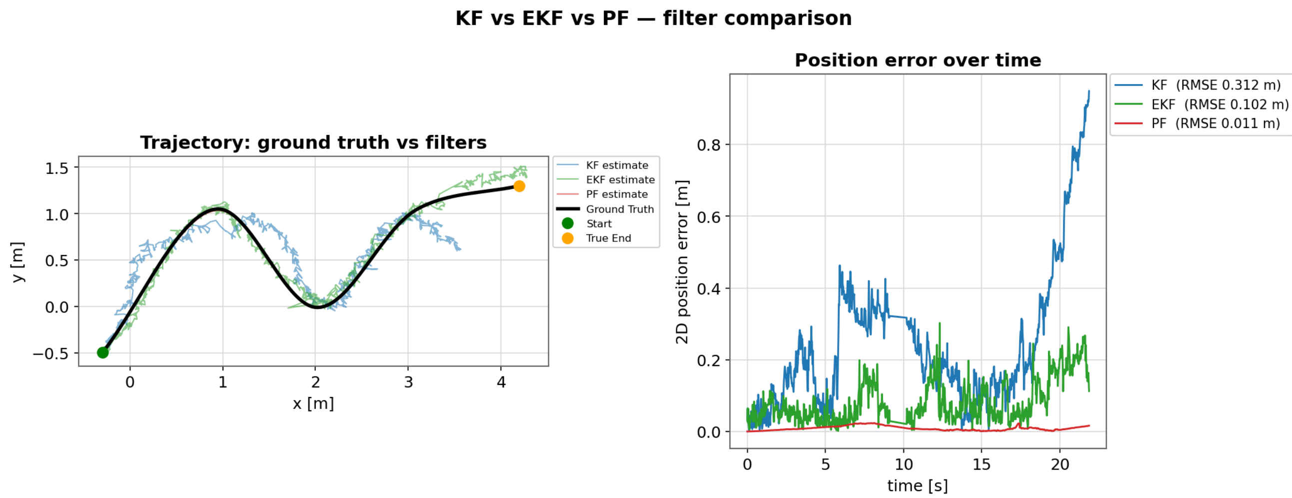

Fig. 2: Left: ground truth versus filter estimates. Right: 2-D position error over time for KF, EKF, and PF with RMSE annotations.

Metric	KF	EKF	PF
Position RMSE [m]	0.312	0.102	**0.011**
Heading RMSE [°]	7.0	2.7	**1.0**
Runtime / step [ms]	0.0054	0.0068	0.149

Tab. 1: Baseline accuracy and runtime on the S-slalom trajectory.

C. EKF Localization Analysis

Fig. 3 decomposes the localization process into three panels. The left panel shows pure dead-reckoning that drifts away from ground truth without corrections. The centre panel adds the growing covariance circles, whose radius expands with travelled distance. The right panel shows the EKF output: with periodic landmark updates the estimate snaps back onto the true track and the covariance stays bounded. The odometry ends $0.89\,\mathrm{m}$ from the true end, whereas the landmark-corrected EKF ends only $0.07\,\mathrm{m}$ away.

D. Effect of Measurement Delay

Fig. 4 and Table 2 show the effect of artificial measurement delay, while Fig. 5 shows the EKF estimate on the slalom at each delay level. RMSE increases monotonically with delay for every filter, because a delayed correction no longer aligns with the current robot position [1]. The PF is the most accurate at zero delay but degrades the most in relative terms, since its tight estimate is the most penalised by the processing lag.

Delay	KF [m]	EKF [m]	PF [m]
0 ms	0.239	0.148	0.005
100 ms	0.261	0.149	0.040
500 ms	0.375	0.238	0.199

Tab. 2: Position RMSE under increasing measurement delay.

E. Process-Noise (Q) and Measurement-Noise (R) Variation

Fig. 6 and Fig. 7 report the Q and R sweeps for the EKF. Both curves are U-shaped, confirming the existence of an optimum. For the process noise, an optimum near $q_{xy} \approx 10^{-5}$ minimises RMSE $(0.149\,\mathrm{m})$: a smaller Q makes the filter over-confident in its biased motion model and it drifts, while a larger Q makes it chase the noisy measurements and become jittery. For the measurement noise, the optimum is near $r \approx 0.05$ $(0.151\,\mathrm{m})$: a smaller R over-trusts the noisy sensor (jitter), while a larger R ignores the sensor and relies on the drifting model. Together these experiments illustrate the model-confidence versus sensor-trust trade-off encoded by Q and R.

V. Summary and Outlook

This paper compared KF, EKF, and PF for unicycle robot localization with a known landmark, on an obstacle-avoiding S-slalom trajectory implemented in ROS 2. The PF achieved the lowest RMSE $(0.011\,\mathrm{m})$, followed by the EKF $(0.102\,\mathrm{m})$ and the KF $(0.312\,\mathrm{m})$; the gap between KF and EKF is explained by the EKF correctly propagating heading uncertainty into position via the motion Jacobian, which the KF omits. Measurement delays of up to $500\,\mathrm{ms}$ degraded all filters monotonically, the PF being most sensitive, and the Q/R sweeps revealed clear U-shaped optima that quantify the model-confidence versus sensor-trust trade-off. The study also surfaced and resolved several practical issues — an initial-pose launch bug, an unbounded trajectory, an EKF linearisation failure near the landmark, and the non-reproducibility of best-effort real-time runs.

Future work should investigate: (1) multiple landmarks and ESS-gated resampling to further improve PF robustness; (2) a 6-DOF model with full IMU pre-integration for 3D environments; and (3) adaptive particle counts to balance accuracy and computational cost.

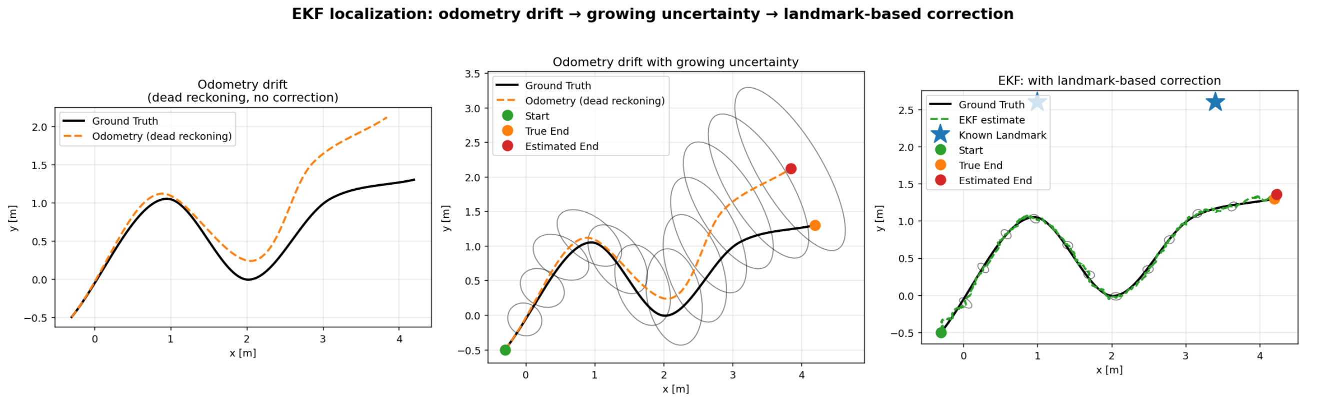

Fig. 3: EKF localization on the S-slalom. Left: odometry drift without corrections. Centre: growing uncertainty (dead-reckoning only). Right: EKF estimate with landmark-based correction; two known landmarks (stars) keep the estimate on the true path.

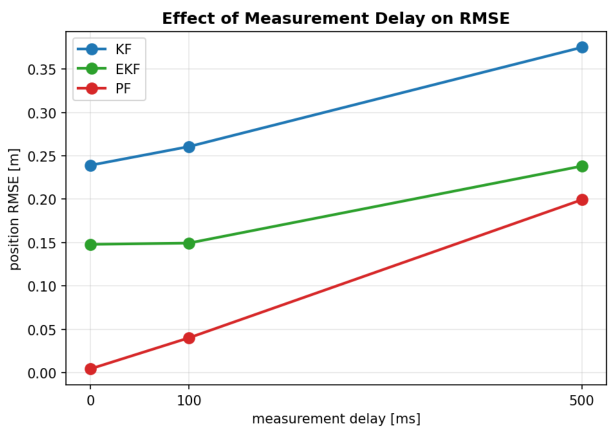

Fig. 4: Position RMSE versus measurement-processing delay for KF, EKF and PF. All filters degrade monotonically with delay.

References

[1] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. New York, NY, USA: Wiley-Interscience, 2001.

[2] F. Dellaert, D. Fox, W. Burgard, and S. Thrun, "Monte carlo localization for mobile robots," in *Proceedings of IEEE International Conference on Robotics and Automation (ICRA)*, 1999, pp. 1322–1328.

[3] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[4] R. C. Smith and P. Cheeseman, "Estimating uncertain spatial relationships in robotics," *Uncertainty in Artificial Intelligence*, vol. 2, pp. 435–461, 1986.

[5] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. Cambridge, MA, USA: MIT Press, 2005.

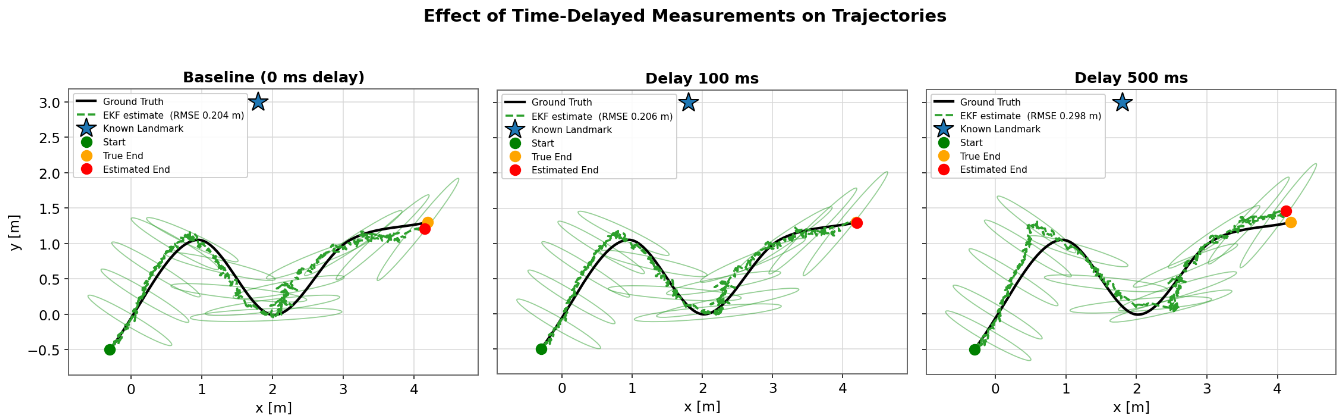

Fig. 5: EKF estimate on the S-slalom under increasing measurement-processing delay (0, 100 and 500 ms): Ground Truth (black), EKF estimate (green dashed) with 2σ covariance ellipses, and the known landmark (blue star). The annotated RMSE (averaged over 30 noise realisations) grows monotonically with delay, as the lagged landmark correction increasingly mismatches the robot's current pose.

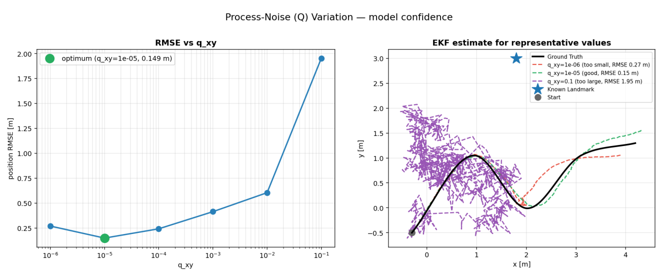

Fig. 6: Process-noise (Q) variation. Left: RMSE versus q_{xy} with the optimum marked. Right: EKF estimate for representative values (too small, good, too large).

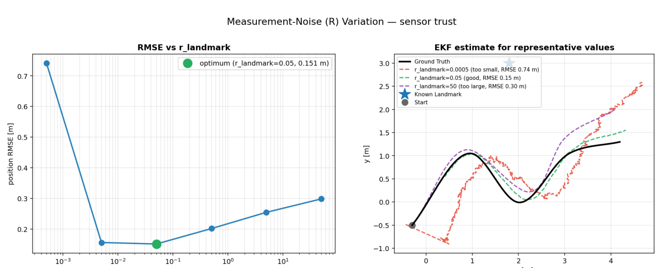

Fig. 7: Measurement-noise (R) variation. Left: RMSE versus r with the optimum marked. Right: EKF estimate for representative values.

Appendix

The complete set of plots and the scripts that generate them are provided in the repository under `prol_filters/scripts` and `data/`. Fig. 8 shows the baseline RMSE as a bar chart for quick reference.

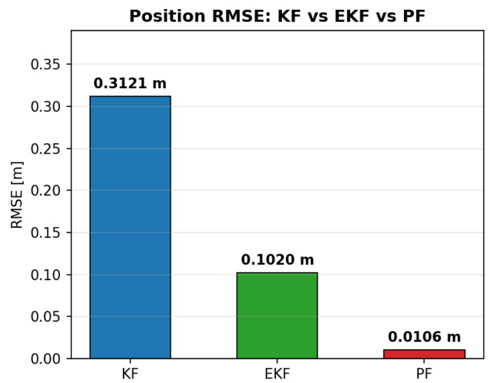

Fig. 8: Baseline position RMSE for KF, EKF and PF on the S-slalom.